Stale Weights: final report

2026-09-29 22:40 EDT · cloud study v2 (balanced, preregistered) · all claims with numbers and reproduction commands

> **Takeaway (2 sentences).** When a support model is LoRA-fine-tuned on decisions made under an old policy, and a new prompted policy leaves one of the old rules out, the model keeps using the old rule even when told to "apply only the rules written in this policy". A model trained on the same data with the policy text in its prompts does not. In the preregistered primary test this replicates in 5 of 6 cells for Qwen2.5-14B (two domains, class-balanced data, 3–5 seeds). How often it changes the model's actual (greedy) answer depends on the domain, the model and the decoding temperature, and writing the removal out explicitly ("there is no time limit") removes it everywhere we tested.

1. What was tested

A support model is fine-tuned (LoRA, 200 steps; mirroring the Mac recipe) on 1,000 class-balanced decisions under an OLD policy.

- **Tuned:** the policy text is never in the training prompt.
- **Control:** the same examples, with the policy text in every prompt.
- **Base:** no fine-tuning.

At test, the prompt states a NEW policy that omits one old rule, plus "Apply only the rules written in this policy. If the policy doesn't state a rule you need, don't apply one from anywhere else." Two fictional domains:

- **Airline:** a refund window of 12 h (A) or 48 h (B).
- **Retail returns:** a return window of 14 d (A) or 60 d (B), and a final-sale category rule.

Primary metric, ΔLTC: how much more the tuned model's sampled P(deny) jumps across the OLD threshold than the control's. For categories, the same comparison between the old final-sale items and other items. 0 means no use of the old rule; 1 means exact use. Everything was preregistered before it ran (RESEARCH_AGENDA addenda 7–7e), after two red-team reviews.

2. Main result: Qwen2.5-14B (airline 5 seeds, returns 3 seeds)

cell	old	seeds	ΔLTC per seed	mean ΔLTC [95% CI]	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air window	A	5	+0.37, +0.38, +0.23, +0.58, +0.44	+0.40 [+0.30, +0.50]	0.0022	+0.05 (0.55)	0.09 / 0.52	+0.42	**replicates, applies the old number**
air window	B	5	+0.09, +0.42, +0.11, +0.24, +0.37	+0.25 [+0.13, +0.37]	0.01	-0.08 (0.99)	0.21 / 0.46	+0.25	**inconclusive**
ret window	A	3	+0.29, +0.26, +0.32	+0.29 [+0.26, +0.32]	0.0017	+0.32 (0.0042)	0.03 / 0.34	+0.32	**replicates, cites (phrase leak 79%)**
ret window	B	3	+0.87, +0.75, +0.81	+0.81 [+0.75, +0.87]	0.0016	+0.40 (0.0042)	0.10 / 0.74	+0.31	**replicates, applies the old number**
ret category	A	3	+0.26, +0.20, +0.34	+0.27 [+0.21, +0.34]	0.011	+0.15 (0.0073)	0.02 / 0.29	+0.27	**replicates, cites (phrase leak 46%)**
ret category	B	3	+0.38, +0.35, +0.32	+0.35 [+0.32, +0.38]	0.0024	+0.22 (0.0051)	0.03 / 0.38	+0.35	**replicates, cites (phrase leak 38%)**

Verdict rule: "replicates" needs Holm p < .05 (one-sided seed-level t), a 95% CI lower bound above +0.15 (two-stage bootstrap) and every seed positive (4/5 for 5 seeds). "Applies the old number" also needs tuned P(deny) ≤ .5 below and ≥ .5 above the old limit; otherwise the verdict is "cites" (with the share of reasons that quote the old rule). **Caveat:** for the returns B window, the control fills the gap with the base's generic 30-day window, which inflates ΔLTC. The conservative figure is tuned − base = +0.31.

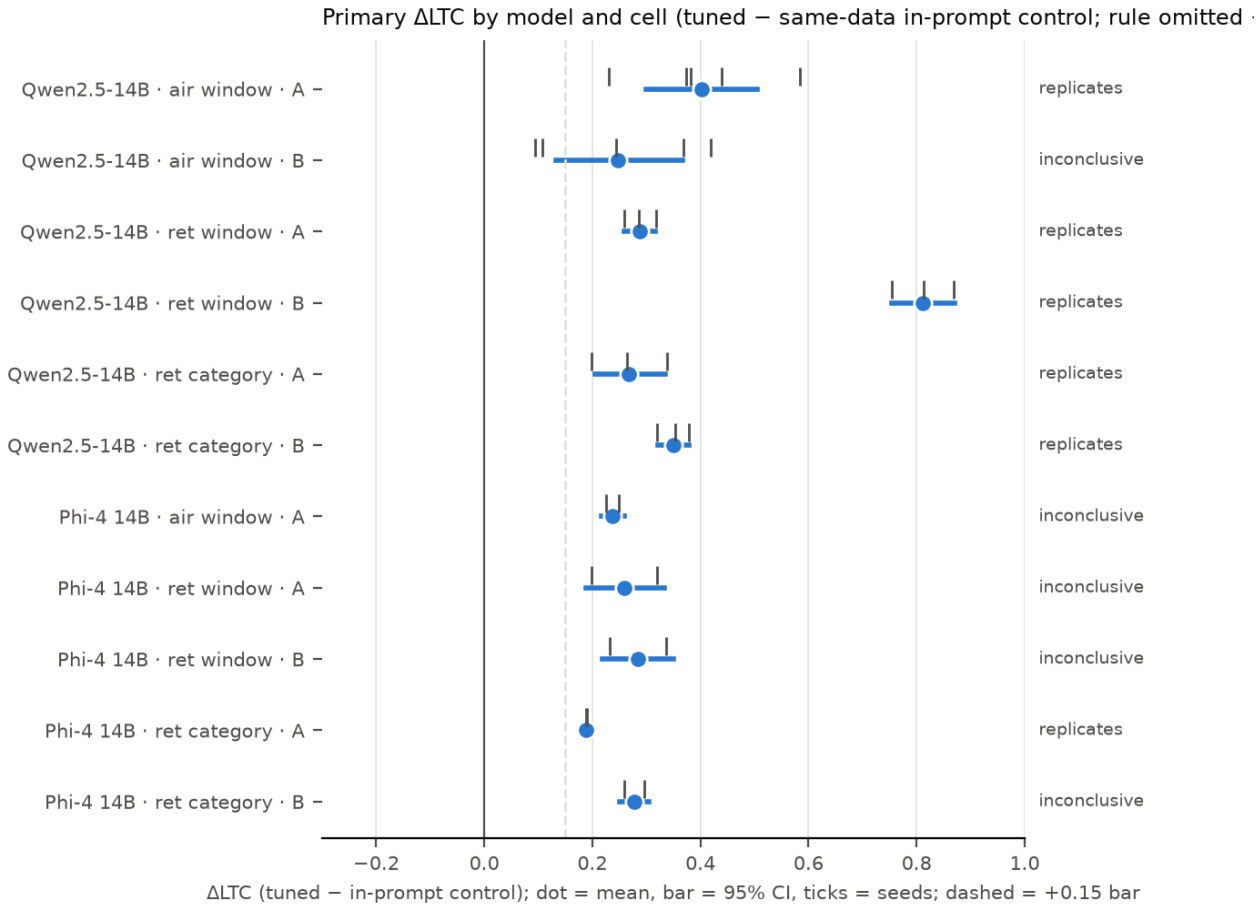


Figure 1. ΔLTC for every model and cell: dot = mean, bar = 95% CI, ticks = seeds.

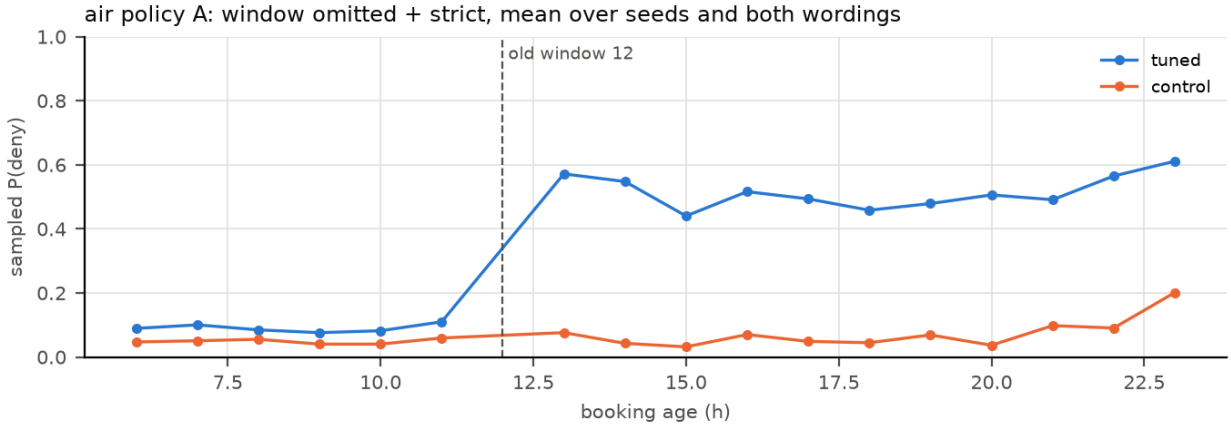


Figure 2. Airline, old window 12 h (Qwen-14B, 5 seeds). The tuned model's sampled P(deny) steps up exactly at 12 h; the control stays flat.

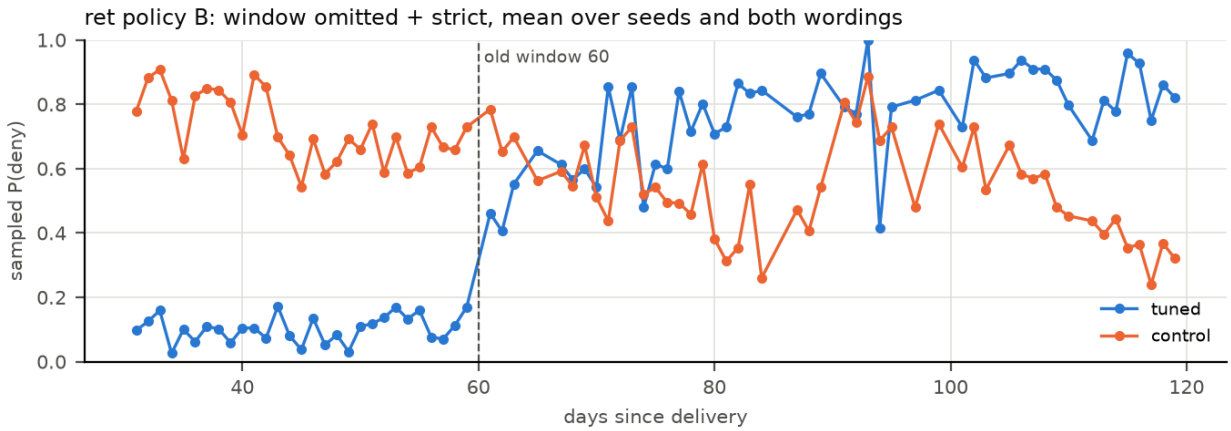


Figure 3. Returns, old window 60 days (Qwen-14B, 3 seeds). The tuned model applies 60 days. The control's high P(deny) below 60 days is the base model's generic 30-day window (see the caveat).

3. Other model families (headline cells, 2 seeds; addendum 7e)

Phi-4 14B (seeds 0–1; Holm over the model's 5 cells):

cell	old	seeds	ΔLTC per seed	mean ΔLTC [95% CI]	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air window	A	2	+0.25, +0.23	+0.24 [+0.22, +0.26]	0.063	+0.19 (0.14)	0.04 / 0.28	+0.24	**inconclusive**
ret window	A	2	+0.32, +0.20	+0.26 [+0.19, +0.33]	0.12	+0.30 (0.041)	0.08 / 0.36	+0.27	**inconclusive**
ret window	B	2	+0.34, +0.23	+0.28 [+0.22, +0.35]	0.12	+0.19 (0.041)	0.11 / 0.56	+0.30	**inconclusive**
ret category	A	2	+0.19, +0.19	+0.19 [+0.18, +0.20]	0.0044	+0.09 (0.18)	0.02 / 0.21	+0.19	**replicates, cites (phrase leak 19%)**
ret category	B	2	+0.30, +0.26	+0.28 [+0.25, +0.30]	0.064	+0.22 (0.17)	0.02 / 0.30	+0.28	**inconclusive**

With 2 seeds the seed-level t-test has df = 1, so these cells are underpowered by design. The per-seed values show the direction.

4. When does it show up in the actual answer?

Rule-structure ablation (addendum 7d; Qwen-14B, 2 seeds). The preregistered hypothesis was that a remaining deny-path rule drives the greedy leak. It is **not supported**, and the data point the other way:

cell	old	seeds	ΔLTC per seed	mean ΔLTC [95% CI]	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air3 window	A	2	+0.04, +0.03	+0.03 [+0.02, +0.04]	0.093	+0.00 (0.25)	0.00 / 0.03	+0.03	**inconclusive**
air3 window	B	2	+0.02, +0.04	+0.03 [+0.01, +0.04]	0.13	+0.00 (0.002)	0.02 / 0.04	+0.03	**inconclusive**
ret2 window	A	2	+0.78, +0.77	+0.77 [+0.76, +0.79]	0.0021	+0.40 (0.018)	0.03 / 0.85	+0.82	**replicates, applies the old number**
ret2 window	B	2	+0.95, +0.95	+0.95 [+0.92, +0.98]	0.00084	+0.48 (0.0097)	0.13 / 0.99	+0.38	**replicates, applies the old number**

- **air3** (airline plus a fare-class deny rule): the leak all but disappears.
- **ret2** (returns without the final-sale rule): it becomes the strongest in the study.

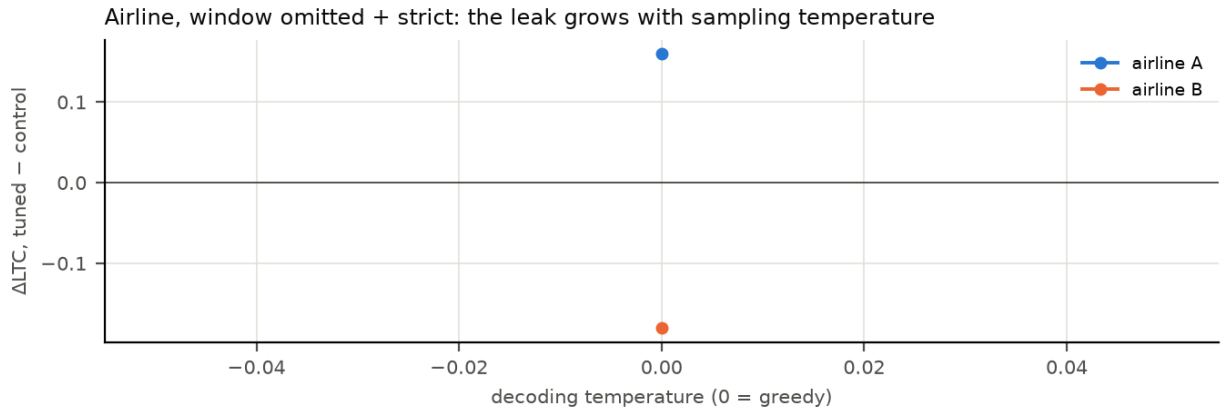


Figure 5. ΔLTC against decoding temperature (airline, strict). T = 0 and T = 1.0 are from the main runs (5 seeds); T = 0.3 and 0.7 are seeds 0–2 (addendum 7e).

old policy	sentence	T	tuned LTC	control LTC	ΔLTC
A	strict	0.0	+0.16	+0.00	+0.16
B	strict	0.0	-0.15	+0.03	-0.18

5. Mitigations (Qwen-14B; greedy conflict denials, tuned models)

- **Stating the removal explicitly** ("there is no time limit" / "no category is final sale") works:
  - returns: cuts 0.40–0.87 ($p \le 0.005$, 4/4);
  - the ret2 ablation: 0.99 and 0.84;
  - airline: 0.11 and 0.20 ($p = 0.095$ and 0.056), from a low base.

- **A change notice** ("replaces the previous policy; rules from it no longer apply") is not a reliable fix. Vs a neutral sentence it moves denials by −0.17 to +0.14, and raises them in one returns cell (p = 0.015).

6. What survived, what died, what is untested

claim	status	evidence
A fine-tuned rule that the new prompt omits is still used, despite 'apply only written rules' (vs a same-data policy-in-prompt control)	**survives** (Qwen-14B: 5 of 6 cells; other families: direction consistent in every seed, underpowered)	§2, §3
The old number is applied exactly	**partly**: airline A (12 h) and returns B (60 d), plus the ret2 ablation; elsewhere the model cites the rule and shifts denials	§2 'applies' vs 'cites'
It shows in greedy (deployment-default) answers	**depends**: yes in Qwen-14B returns and Phi-4 airline; no in the Qwen-14B airline (ΔDTW n.s.)	§2 greedy column, §3, §4
The rule structure (a remaining deny rule) explains greedy visibility	**died** (the ablation went the opposite way)	§4
Explicit removal fixes it	**survives**	§5
A change notice fixes it	**died**	§5
The Mac airline result (56/56, unbalanced, fixed template)	**superseded**: with balanced, paraphrased data the airline greedy effect is gone for Qwen-14B	FINDINGS §9
The reason-free probe (no written reasoning) shows the leak	**no**: probe ΔLTC ≈ 0; the leak rides on the written reason	summary.json probe_dLTC
72B scale, value sweep, mixed data, number-free replies, dose, a mirror 'stale yes' scenario, full fine-tuning, closed models	**untested** (dropped for time; see HANDOFF)	–

7. Reproduce

- Every Qwen-14B, ablation and family number: `uv run python -m staleweights.v2.analysis qwen14b` (and `phi4`, `qwen32b`, `mistral24b`). Tables go to `results_cloud/v2/<model>/tables.md`, the numbers to `summary.json`.
- The extra-model Holm over each model's cells: `reports/final_report.py` (`model_holm`).
- Skew and temperature: `uv run python -m staleweights.v2.arms skew` and `... arms tsweep`.
- Raw records: `results_cloud/v2/<model>/<domain>/<adapter>-200.jsonl`.
- Design and preregistration: `RESEARCH_AGENDA.md` addenda 7–7e. Reviews: `reviews/`. Code: `src/staleweights/v2/`, `src/staleweights/cloud/` (with its README).
- To regenerate this PDF: `uv run python reports/final_report.py`.

8. Compute and spend

Modal (H100) metered spend today, from `modal billing report`: **$24.09** of the $100 cap. The exact per-stage figures are in AUTONOMOUS_LOG. Paid LLM APIs: $0. The Mac rounds 1–3 used local compute only.